# Smart Contract Audit & Deployment Ledger | My Robinhood Airdrop Project

**Project Home：** https://github.com/shin-kado/My-Robinhood-Airdrop-Project

### Technical Environment

| | |
|---|---|
| Network Name | Robinhood Chain Testnet |
| Chain ID | 46630 |
| License | MIT |
| Solidity Version | 0.8.20 (Source Code Level) |
| Compiler Version | 0.8.31+commit.fd3a2265 |
| EVM version | Default (osaka) |
| Optimization | No |

### Development Tools

| | |
|---|---|
| IDE / Editor | VS Code (Local) linked with Remix IDE via remixd |
| Wallet | MetaMask |
| Explorer | Robinhood Chain Testnet Explorer |

### Web3 Portal

| | |
|---|---|
| URL | https://shin-kado.github.io/My-Robinhood-Airdrop-Project/web-app/ |

### MyToken

| | |
|---|---|
| Source Code | Mytoken.sol |
| Interface | Remix IDE / Blockscout (Verified Explorer) |
| Function / Features | A foundational digital asset issuance module for building a project-specific ecosystem. Fully compliant with the industry-standard ERC-20, it ensures high security and interoperability. It generates a predefined supply upon deployment and serves as the primary asset for various distribution and time-lock experiments.<br>（プロジェクト独自の経済圏を構築するための、基礎となるデジタル資産発行モジュールです 。業界標準のERC-20規格に準拠し、高い安全性と互換性を備えています。デプロイと同時に 特　定の数量を生成し、その後の「一括配布」や「タイムロック」などの実証実験における主要な配布資産として機能します。） |
| Contract Address | 0x196eCa072F41571233E4F6D215F89A3446DD569b |
| Standard / 規格 | ERC-20 (OpenZeppelin準拠) |
| Token Name | My Robinhood Token |
| Symbol | MRT |
| Total Supply / 供給量 | 1,000,000 MRT |
| Decimals / 小数点桁数 | 18 (Ethereum Standard) |
| Status | Verified (Partial Match) |
| Verified Explorer | https://explorer.testnet.chain.robinhood.com/address/0x196eCa072F41571233E4F6D215F89A3446DD569b?tab=contract |
| Transaction Hash | 0xe0471842f2f1c465f678a835f83a4cb961de86b964e12b79795dd8faca1c8e9f |

### Persistent On-chain Activity Registry / 定期的オンチェーン・アクティビティ記録システム

| | |
|---|---|
| Source Code | Storage.sol |
| Interface | Remix IDE / Blockscout (Verified Explorer) |
| Function / Features | A system designed to prove and accumulate consistent blockchain usage records (on-chain activity) by periodically registering date data.<br>（毎週定期的に日付データを記録することで、ブロックチェーン上での継続的な利用実績（アクティビティ）を証明・蓄積するためのシステムです。） |
| Contract Address | 0x14cbB4da6Ad42877d065eE1D1398c984E0b1a426 |
| Status | Verified (Partial Match) |
| Verified Explorer | https://explorer.testnet.chain.robinhood.com/address/0x14cbB4da6Ad42877d065eE1D1398c984E0b1a426?tab=contract |

### Secure Authority Management System / 管理者権限セキュア管理

| | |
|---|---|
| Source Code | Owner_Simple.sol |
| Interface | Remix IDE / Blockscout (Verified Explorer) |
| Function / Features | A management system that defines a contract "Owner," restricting critical operations to authorized users and enabling the secure transfer of administrative privileges.<br>（スマートコントラクトに「管理者（Owner）」を設定し、重要な操作を許可された人だけに制限したり、その権限を安全に引き継いだりするための管理システムです。） |
| Contract Address | 0xca18d49df82e420BBd40a46c95A48d1717976Dd1 |
| Status | Verified (Partial Match) |
| Verified Explorer | https://explorer.testnet.chain.robinhood.com/address/0xca18d49df82e420BBd40a46c95A48d1717976Dd1?tab=contract |

### Transparent On-chain Voting & Delegation / 透明型オンチェーン投票・委任管理

| | |
|---|---|
| Source Code | Ballot.sol |
| Interface | Remix IDE / Blockscout (Verified Explorer) |
| Function / Features | A management system for securely executing transparent on-chain voting and delegating voting rights to trusted third parties, enabling tamper-proof tallying and winner selection.<br>（ブロックチェーン上での透明性の高い直接投票に加え、信頼できる第三者への投票権委任（デリゲーション）を安全に実行し、改ざん不能な形式で集計・選出を行うための管理システムです 。） |
| Contract Address | 0x9f4801277A2e2C67896A06E89021e6bcef7c9106 |
| Status | Verified (Partial Match) |
| Verified Explorer | https://explorer.testnet.chain.robinhood.com/address/0x9f4801277A2e2C67896A06E89021e6bcef7c9106?tab=contract |
| Transaction Hash(Give right to Vote) | 0xd8d2c4570eb75edbbec3faa92277d19277b8f242d24bb4d58947b1712c40c9e5 |
| Transaction Hash(Vote) | 0x31d5d543ba91dd2e487ce1f848efd0e02f3dba47e44ca10c325f7fed4349b675 |

# Smart Contract Audit & Deployment Ledger | My Robinhood Airdrop Project

### Standard Batch Token Distribution / トークン一律一括配布

| | |
|---|---|
| Source Code | SimpleAirDrop.sol |
| Interface | Remix IDE / Blockscout (Verified Explorer) |
| Function / Features | A specialized tool for batch-transferring a fixed amount of ERC20 tokens to multiple addresses. It streamlines the process of delivering tokens to many users under identical conditions, such as event rewards or commemorative token distributions.<br>（指定したERC20トークンを、複数のアドレスへ「一律の数量」で一括送金するための専用ツールです。イベントの参加報酬や記念トークンの配布など、多数のユーザーへ同一条件でトークンを届ける操作を、安全かつ最小限の手間で実行できます。） |
| Contract Address | 0x1926e8C311Ae38Bd2d20A2755852f8B5aF0F4258 |
| Status | Verified (Partial Match) |
| Transaction Hash(Approve) | https://explorer.testnet.chain.robinhood.com/address/0x1926e8C311Ae38Bd2d20A2755852f8B5aF0F4258?tab=contract |
| Transaction Hash(Approve) | 0xf52f53670303735f0adbaef9aec12776ccd31b95a7faefd79f45f446139147bf |
| Transaction Hash(sendTokens) | 0x97b8b316a0bf42bd3b2e5910a7c3082fe3730d224cf9ff1fb81cf50c2901dd37 |

### Advanced Multi-Token Distribution / 次世代型・マルチトークン一括配布

| | |
|---|---|
| Source Code | AdvancedAirdrop.sol |
| Interface | Remix IDE / Blockscout (Verified Explorer) |
| Function / Features | A centralized system for the efficient batch distribution of custom and equity tokens to multiple addresses. It features both "Standard" (fixed amount) and "Advanced" (variable amount) distribution modes to streamline large-scale airdrop operations.<br>（独自のカスタムトークンや株式トークンを、複数のアドレスへ一括で効率的に配布するためのシステムです。全員に同額を配る「標準配布」と、宛先ごとに異なる数量を指定できる「高度配布」の2つの機能を備え、大規模なエアドロップ操作を簡略化します。） |
| Contract Address | 0x86238ed61A026604a58D43C554Bde90281f05102 |
| Status | Verified (Partial Match) |
| Verified Explorer | https://explorer.testnet.chain.robinhood.com/address/0x86238ed61A026604a58D43C554Bde90281f05102?tab=contract |
| Transaction Hash(Approve) | 0x0a64943ba048f0048dc36e547eb24c024ae24723b225b1c1b083f89027bab3ea |
| Transaction Hash(sendTokens) | 0x831e7fce6bfc8fb535a0e8707d5ee903c0bc3c629df41f7e078eed6a93862f01 |

### Time-Locked On-chain Asset Protection / 時間制限付きオンチェーン・資産保護

| | |
|---|---|
| Source Code | TimeLockAirdrop.sol |
| Interface | Remix IDE / Blockscout (Verified Explorer) |
| Function / Features | An advanced management system that implements "Time Locks" on token distributions, restricting withdrawals until a specified date and time. It enforces strategic asset management, such as scheduled token releases or long-term incentive distributions, through automated smart contract logic.<br>（トークン配布において「時間制限（ロック）」を設け、指定した日時が経過するまで引き出しを制限する高度な管理システムです。計画的なトークン放出や、長期保有を前提とした報酬配布など、戦略的な資産運用をスマートコントラクトによって自動強制します。） |
| Contract Address | 0x9258F07C118e2F141c676d330A70eAf630b2F90E |
| Status | Verified (Partial Match) |
| Verified Explorer | https://explorer.testnet.chain.robinhood.com/address/0x9258F07C118e2F141c676d330A70eAf630b2F90E?tab=contract |
| Transaction Hash(Approve) | 0x700d91af67722b65e4d5c6b98ab1ec96ace47ef0497eb098a0e6935e0c88ae4a |
| Transaction Hash(sendTokens) | 0xa6098e5a1780c23630e95d1e1e0312885c9ff0ece931369616ae18978df791ad |
| Transaction Hash(ClaimTokens) | 0x6e0b4843f8f8a64d94ae7839dffcf3bcff06c5fbceb391343963015b22b8e745 |

### Batch Distribution Module with Circuit Breaker / 緊急停止機能付き一括配布

| | |
|---|---|
| Source Code | SimpleAirDropPausable.sol |
| Interface | Remix IDE / Blockscout (Verified Explorer) |
| Function / Features | A distribution control module that integrates an emergency "Circuit Breaker" into batch token transfer functionality. It allows administrators to instantly lock all operations in response to errors or unforeseen events, providing a robust mechanism to prevent asset leakage or misdirection.<br>（複数のアドレスへトークンを一括配布する機能に、管理者による「緊急停止スイッチ（サーキットブレーカー）」を組み込んだ制御モジュールです。誤設定や不測の事態が発生した際に、即座にすべての配布動作をロックし、資産の流出や誤送信を未然に防ぐ仕組みを提供します。） |
| Contract Address | 0x1cd4DB04175bfdD2e8802d52C995B79294Dc5ac8 |
| Status | Verified (Partial Match) |
| Verified Explorer | https://explorer.testnet.chain.robinhood.com/address/0x1cd4DB04175bfdD2e8802d52C995B79294Dc5ac8?tab=contract |
| Transaction Hash(Approve) | 0x38cb4da2a2115787fd8edbd3766ea7ce83030d7fed7a230035301810d607e8c4 |
| Transaction Hash(setPaused) | 0xbd44bbd48a44ab0b796b8210fd5d53f837034b9077e3f724806670df48f01368 |
| Transaction Hash(sendTokens Failed) | 0x13a754d00a83f2509429e30823c8a999a4506dd6c5e1f938d33cee748a8b37c4 |
| Transaction Hash(setPaused) | 0xda1099ec203acaeab2ecc4981fbcaed8084b6df2a323f9a8277b0aacb868e08c |
| Transaction Hash(sendTokens Success) | 0xc4e8c199dce1f0d54661f47ead4769e236bf4eb389b4eb4d39781b5c08d5ee45 |

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### Multi-Asset Batch Transfer Exchanger / 多種類トークン一括配布

| | |
|---|---|
| Source Code | MultiTokenTransfer.sol |
| Interface | Remix IDE / Blockscout (Verified Explorer) |
| Function / Features | An execution module designed to dispatch multiple types of tokens to a designated address in a single operation. It streamlines the transfer process, significantly reducing manual effort and gas fees.<br>（種類の異なる複数のトークンを、一回の手続きで指定のアドレスへまとめて発送するための実行モジュールです。送金作業を一本化することで、手間とガス代（手数料）を大幅に削減する仕組みを提供します。） |
| Contract Address | 0xa547a0B7c0CA10fcf9E904eB9656942E3578e1D7 |
| Status | Verified (Partial Match) |
| Verified Explorer | https://explorer.testnet.chain.robinhood.com/address/0xa547a0B7c0CA10fcf9E904eB9656942E3578e1D7?tab=contract |
| Transaction Hash(Approve1) | 0xf9ebec67c155215fe40583673f3ea52f43f081745f45308db0d1341953b5593d |
| Transaction Hash(Approve2) | 0x225dc7b424ddaca6520bd82418030f5145b4f7981d300b8bafe29b8bd92a376f |
| Transaction Hash(Approve3) | 0x7c8950a8ee520e44ad0fd2b6008382770e27e2c803ab327d1df836ca386eb2f9 |
| Transaction Hash(sendTokens) | 0x5b805ea234be395dfbba7408d54ff589c1a0db300d8fff50a7d049375f832188 |

### Authorized Multi-Token Batch Dispatch / 権限認証型・多種類トークン一括配布

| | |
|---|---|
| Source Code | MultiTokenTransferWithWhitelist.sol |
| Interface | Remix IDE / Blockscout (Verified Explorer) |
| Function / Features | A batch distribution module that combines multi-token transfer functionality with a "Whitelist" authentication mechanism. It restricts operations to pre-authorized addresses only, ensuring a secure distribution process managed by verified operators.<br>（複数の異なるトークンを一括送付する機能に、特定の承認済みアドレスのみが実行できる「ホワイトリスト制」を導入した制御モジュールです。不特定多数による操作を制限し、あらかじめ許可された運用者のみが資産移動を行える、セキュアな一括配送の仕組みを提供します。） |
| Contract Address | 0xEC4Ae1A488a4615d5e159c6c0675c38a9Bd27376 |
| Status | Verified (Partial Match) |
| Verified Explorer | https://explorer.testnet.chain.robinhood.com/address/0xEC4Ae1A488a4615d5e159c6c0675c38a9Bd27376?tab=contract |
| Transaction Hash(Approve) | 0x84f13053b7697d806d85753bc357b2ed81d0ef93612d3160a14365993d0ed7e8 |
| Transaction Hash(Update Whitelist True) | 0x59442f2b4172998a98b3d6e0419b75ef5622c73539d424a59d78c792e46fde88 |
| Transaction Hash(sendTokens) | 0xa3541feb75b47f6638a3aab6eca474f2f1b7d04876c3e5bddcd9eef1dc28fc15 |
| Transaction Hash(Approve) | 0x8d3f975bc42cc374fda6ee631db328273d640a7f1062e39ed053b757b83a48db |
| Transaction Hash(Update Whitelist False) | 0x1cca605cfce9f2ab62e8c65128ace69954b032f79758d4cead452c4ecb05149a |
| Transaction Hash(sendTokens) | 0x5d76d4d3ac638bbd552eb2034d0b26ab7ed9f1b43f5f483e618307ea183722c1 |

### Time-Limited On-chain Voting Secure Management / 時間制限付きオンチェーン投票セキュア管理

| | |
|---|---|
| Source Code | TimedBallot.sol |
| Interface | Remix IDE / Blockscout (Verified Explorer) |
| Function / Features | A secure voting system that introduces a "Time Limit" to standard electronic voting processes. It automatically closes voting after a predefined deadline, ensuring a fair and transparent decision-making process through strict on-chain temporal management.<br>（標準的な電子投票機能に「時間制限」の概念を導入したシステムです。デプロイ時に設定された期限を過ぎると自動的に投票が締め切られ、ブロックチェーンのタイムスタンプを利用した厳格な期間管理により、公正な意思決定をサポートします。） |
| Contract Address | 0x52798A5079e9BCf4409D7287a88bA05b1d72D6a0 |
| Status | Verified (Partial Match) |
| Verified Explorer | https://explorer.testnet.chain.robinhood.com/address/0x52798A5079e9BCf4409D7287a88bA05b1d72D6a0?tab=contract |
| Transaction Hash(Give right to Vote) | 0x868f1f1b837ccf4b0e1ec4b56aab098ed2e9ee3edcc4e328b720b88d909c3c7a |
| Transaction Hash(Vote Success) | 0x0cb22515ada87dd367fbdf99db5d7a6ada8d325799de582cec9d74b68796fd76 |
| Transaction Hash(Vote Failed) | 0x36a03456268cbb8421a6df951d214c85d19904f6ccaa27c31e81858e76b719af |

### On-chain Achievement & Digital Badge Minting Engine / オンチェーン実績証明・デジタルバッジ発行エンジン

| | |
|---|---|
| Source Code | MyNFT.sol |
| Interface | Remix IDE / Blockscout (Verified Explorer) |
| Function / Features | A module for issuing tamper-proof achievement certificates (NFTs) as proof of completing specific tasks. It securely links metadata to the blockchain, providing a mechanism to officially verify the holder's skills and contributions as digital assets.<br>（特定の工程を完了した証として、改ざん不能な実績証明（NFT）を発行するためのモジュールです。メタデータ（画像や説明文）をブロックチェーンに強固に紐付け、保有者のスキルや貢献をデジタル資産として公式に証明する仕組みを提供します。） |
| Contract Address | 0x680AeA01Ee91170d1927Cccb6DA0067875D1e1fC |
| Status | Verified (Partial Match) |
| Verified Explorer | https://explorer.testnet.chain.robinhood.com/address/0x680AeA01Ee91170d1927Cccb6DA0067875D1e1fC?tab=contract |
| URL(JSON) | https://raw.githubusercontent.com/shin-kado/My-Robinhood-Airdrop-Project/refs/heads/main/docs/badge.json |
| Transaction Hash(Mint NFT) | 0x921e132786ec489400660e6478d0e2419413291cf3a4eb3da4020071639da4e8 |

# Smart Contract Audit & Deployment Ledger | My Robinhood Airdrop Project

### Blockchain Basic Data Storage Module / ブロックチェーン・ベーシックデータ記録モジュール

| | |
|---|---|
| Source Code | SimpleStorage.sol |
| Interface | Remix IDE / Blockscout (Verified Explorer) |
| Function / Features | A foundational data management module designed to securely record and retrieve a single numerical value on the blockchain. It provides the simplest mechanism for data storage in Web3, enabling information updates and persistence across a decentralized network without a central server.<br>（ブロックチェーン上に単一の数値を安全に記録し、いつでも正確に呼び出すことができる基礎的なデータ管理モジュールです。サーバーを介さずに、分散型ネットワーク上で情報の書き換えと保持を行う、Web3における最もシンプルなデータ保存の仕組みを提供します。） |
| Contract Address | 0x299396e27CDd54C20EEA8BE61E5881ee851d1FeD |
| Status | Verified (Partial Match) |
| Verified Explorer | https://explorer.testnet.chain.robinhood.com/address/0x299396e27CDd54C20EEA8BE61E5881ee851d1FeD |
| Transaction Hash(Set) | 0x5f0cc8f9d58a6794caca5e1d155872cb94eef9feede6288361ed6f1b8a9bebc6 |

### Advanced On-chain Multi-Data Permissioned Registry / 次世代型ブロックチェーン・データ記録レジストリ

| | |
|---|---|
| Source Code | advanced-storage.html / AdvancedStorage.sol |
| Interface | Web Browser Portal / Remix IDE / Blockscout (Verified Explorer) |
| Contract Address | 0x719D976CA63f1491F616a1bC11784c0372CD6262 |
| Function / Features | A robust utility for securely storing various string and numeric data on-chain, featuring advanced management based on owner-level permissions.<br>（文字列や数値などの多様なデータをブロックチェーン上に安全に保存し、所有者権限に基づいた高度なデータ管理を実現します。） |
| Status | Verified (Partial Match) |
| Verified Explorer | https://explorer.testnet.chain.robinhood.com/address/0x7212dD5F5a2062a412dc3441A1571A2c335CD7fF |
| Transaction Hash(Set Num Success) | 0xffbe58d951245a420f910c3fb9e0319a7f465893b50fa4742bb8ef4ecae24b1e |
| Transaction Hash(Set Text Success) | 0xc32239193f41b811dbac0235571674a7ed425ff7c0d61954bf2d01ff5d10efc5 |
| Transaction Hash(Owner Set Num Success) | 0xc053eb58e48d34e640491eb26a0d9cafb0e74a109d5e1512edd23fe984ee3bae |
| Transaction Hash(Owner Set Text Success) | 0xf050ae87da2869ef6718f2ed78c0bdb22201ed325cb52821af3b71d9d7e6299d |
| Transaction Hash(Set Num Success) | 0x423ce9b8e9e5e78127d563d9bc6d95118cbcc9ebb68fe78e3213d9aad194d8b2 |
| Transaction Hash(Set Text Success) | 0x7b078ef4c2d0f9457be3dbbd065cc3b0308e33231748fb785e4966a5aab74db0 |
| Transaction Hash(Owner Set Num Failed) | 0xd799b1dc08c28cefdc50a3c62f6936f6be0c27623110f312c7c7ee92145da12c |
| Transaction Hash(Owner Set Text Failed) | 0xc74fd1424e0cb95779955dfe0e71dfc9360469db98e8dbb4ef4032898c6f6ddf |
| URL | https://shin-kado.github.io/My-Robinhood-Airdrop-Project/web-app/projects/advanced-storage.html |

### Web3 Asset & Token Distribution Dashboard / Web3資産＆トークン配布ダッシュボード

| | |
|---|---|
| Source Code | dashboard.html / tokens.js / accounts.js / colors.js |
| Interface | Web Browser Portal |
| Function / Features | A visualization module for the integrated management of on-chain assets and token distribution history. It transforms complex blockchain data into an intuitive dashboard, enabling real-time monitoring and transparent verification of activities.<br>（分散型ネットワーク上の資産状況とトークン配布の実績を、グラフィカルに統合管理するための表示モジュールです。複雑なオンチェーンデータを直感的なダッシュボードに変換し、リアルタイムでの資産モニタリングと透明性の高い実績確認を可能にします。） |
| Interface | Remix IDE / 各種ダッシュボード |
| URL | https://shin-kado.github.io/My-Robinhood-Airdrop-Project/web-app/projects/dashboard.html |